

Conflict-Aware User Scheduling and Hybrid Beamforming for Near-Field STAR-RIS Systems

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Abstract—This paper investigates conflict-aware user scheduling and hybrid beamforming for near-field simultaneously transmitting and reflecting reconfigurable intelligent surface (STAR-RIS)-assisted multi-user millimeter-wave systems. In dense user scenarios, conventional gain-based scheduling often suffers from severe beam conflicts and poor fairness, whereas orthogonality-based schemes improve fairness at the cost of throughput. To address this issue, we propose a conflict-aware semi-orthogonal user scheduling and hybrid beamforming framework, termed CA-SUS-HBF, which jointly captures beam conflict, channel orthogonality, and user-rate balance. In addition, a validation-driven tradeoff tuning mechanism is introduced to balance sum-rate and fairness in a principled manner. Simulation results show that the proposed method achieves a favorable throughput-fairness tradeoff over conventional top-gain and semi-orthogonal baselines, while maintaining competitive sum-rate performance close to the fully digital near-field upper bound in practical operating regimes.

Index Terms—STAR-RIS, near-field communications, hybrid beamforming, user scheduling, conflict-aware.

I. INTRODUCTION

Reconfigurable intelligent surfaces (RIS) have recently attracted considerable attention as a promising technology for improving spectral efficiency and coverage by intelligently reconfiguring the wireless propagation environment [1], [2]. By employing many controllable passive elements, RIS can reshape the propagation field and create more favorable transmission conditions without requiring a large number of active radio-frequency components. As an important extension, simultaneously transmitting and reflecting RIS (STAR-RIS) enables full-space coverage by serving users located on both sides of the surface, thereby offering greater flexibility than conventional reflecting-only RIS in dense multi-user deployments [3], [4]. Meanwhile, millimeter-wave (mmWave) systems rely on large antenna arrays and directional beamforming to compensate for severe path loss, and hybrid beamforming (HBF) has become a practical architecture for achieving a favorable tradeoff between hardware complexity and beamforming performance under limited RF-chain constraints [5], [6]. In such systems, user scheduling plays a critical role because the number of candidate users is typically larger than the number of simultaneously supportable streams.

Conventional user scheduling methods mainly include gain-based selection and semi-orthogonal user selection (SUS) [7],

[8]. Gain-based methods prefer users with strong effective channels and may provide good throughput in lightly loaded scenarios, but they do not explicitly mitigate the strong inter-user interference caused by overlapping beams. In contrast, SUS aims to reduce channel correlation among scheduled users [9], yet it may sacrifice useful beamforming gain by excluding strong but moderately correlated users. As a result, these two classes of schedulers often emphasize different objectives, namely instantaneous throughput and spatial separability, without directly controlling the tradeoff between them. This issue becomes even more critical when the candidate-user pool becomes large, since a larger pool simultaneously brings stronger multi-user diversity and a higher risk of beam collision.

These limitations become more severe in near-field STAR-RIS systems, where the effective channel depends not only on angular direction but also on spatial focusing characteristics. Unlike conventional far-field beamforming, near-field transmission exhibits distance-dependent phase variations and focusing effects, so users with similar angular directions may still experience different channel structures, while users in different positions may nevertheless compete for highly correlated analog beams. As a result, multiple users may contend for similar beam resources, which creates serious beam conflicts in dense user scenarios and complicates the design of efficient HBF. Therefore, a scheduling strategy for near-field STAR-RIS systems should jointly account for channel strength, orthogonality, and beam-conflict behavior.

Although a number of prior works have investigated active and passive beamforming optimization, resource allocation, and protocol design for RIS- and STAR-RIS-assisted systems [3], [4], [10], the role of beam conflict in user scheduling for near-field STAR-RIS-aided HBF systems remains insufficiently explored. In particular, most existing studies focus on beamforming or passive surface optimization itself, while user scheduling is either simplified or treated separately from analog beam assignment. Consequently, there is still a lack of low-complexity scheduling methods that jointly account for effective channel gain, beam conflict tendency, spatial separability, and user-rate balance in a unified manner.

Motivated by this gap, this paper proposes a conflict-aware semi-orthogonal user scheduling and HBF framework, termed CA-SUS-HBF, for near-field mmWave STAR-RIS systems. The proposed method evaluates candidate users using a

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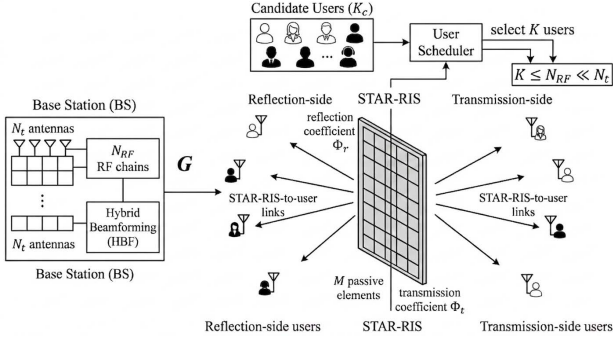


Figure 1. Near-field STAR-RIS-assisted multi-user mmWave system with HBF.

conflict-aware priority metric that captures effective channel strength, beam conflict tendency, and channel orthogonality. Based on this metric, an initial scheduled-user set is constructed and then refined jointly with analog beam assignment and digital precoding. To balance throughput and fairness, a utility function combining spectral-efficiency and fairness-related terms is adopted, and its parameters are selected through a validation-based tuning procedure rather than ad hoc choice. Simulation results show that the proposed scheme achieves a more favorable throughput–fairness tradeoff than conventional top-gain and semi-orthogonal baselines, while maintaining sum-rate performance close to a fully digital near-field upper bound in practical operating regimes. In particular, the proposed design is especially effective in user-clustered scenarios, where beam conflict becomes a dominant factor limiting the performance of conventional scheduling strategies.

II. SYSTEM MODEL

As illustrated in Fig. 1, we consider a downlink near-field STAR-RIS-assisted multi-user millimeter-wave system, where both the BS–STAR-RIS link and the STAR-RIS–user links are modeled under spherical-wave propagation. A base station (BS) equipped with N_t antennas and N_{RF} RF chains serves multiple single-antenna users. A STAR-RIS with M passive elements is deployed to enhance the wireless coverage. Let $\mathcal{K}_c = \{1, 2, \dots, K_c\}$ denote the candidate-user set, from which K users are scheduled for transmission, where $K \leq N_{RF} \leq N_t$.

The BS employs HBF composed of an analog precoder $\mathbf{F}_{RF} \in \mathbb{C}^{N_t \times N_{RF}}$ and a digital baseband precoder $\mathbf{F}_{BB} \in \mathbb{C}^{N_{RF} \times K}$. The transmitted signal is given by $\mathbf{x} = \mathbf{F}_{RF} \mathbf{F}_{BB} \mathbf{s}$, where $\mathbf{s} \in \mathbb{C}^{K \times 1}$ is the data symbol vector satisfying $\mathbb{E}\{\mathbf{s}\mathbf{s}^H\} = \mathbf{I}_K$. The total transmit power constraint is

$$\|\mathbf{F}_{RF} \mathbf{F}_{BB}\|_F^2 \leq P_{\max}, \quad (1)$$

where $P_{\max}$ is the total transmit power.

The STAR-RIS simultaneously serves users located in the transmission and reflection regions. Let $\mathcal{K}_T$ and $\mathcal{K}_R$ denote the sets of users in the transmission and reflection regions,

respectively, with $\mathcal{K}_T \cup \mathcal{K}_R = \mathcal{K}_c$ and $\mathcal{K}_T \cap \mathcal{K}_R = \emptyset$. The STAR-RIS response matrix is expressed as

$$\Phi_s = \text{diag}(\beta_{1,s} e^{j\phi_{1,s}}, \beta_{2,s} e^{j\phi_{2,s}}, \dots, \beta_{M,s} e^{j\phi_{M,s}}), \quad (2)$$

for $s \in \{T, R\}$, where $\beta_{m,s} \in [0, 1]$ and $\phi_{m,s} \in [0, 2\pi)$ denote the amplitude and phase coefficients of the m -th STAR-RIS element for region s . Under the energy-splitting protocol, the coefficients satisfy

$$\beta_{m,T}^2 + \beta_{m,R}^2 \leq 1, \quad \forall m = 1, \dots, M. \quad (3)$$

For a scheduled user k , the effective cascaded channel includes the direct BS–user path and the STAR-RIS-assisted path, i.e.,

$$\tilde{\mathbf{h}}_k^H = \mathbf{h}_{BU,k}^H + \mathbf{h}_{SU,k}^H \Phi_{s_k} \mathbf{H}_{BS}, \quad (4)$$

where $\mathbf{h}_{BU,k} \in \mathbb{C}^{N_t \times 1}$ is the direct BS–user channel, $\mathbf{H}_{BS} \in \mathbb{C}^{M \times N_t}$ is the BS–STAR-RIS channel, $\mathbf{h}_{SU,k} \in \mathbb{C}^{M \times 1}$ is the STAR-RIS–user channel, and $s_k \in \{T, R\}$ indicates whether user k belongs to the transmission or reflection side.

Unlike conventional far-field models that depend mainly on angular information, the considered near-field channels are distance-dependent and follow a spherical-wave propagation structure. Specifically, the array response is determined by the absolute propagation distance between the source point and each array element. For example, the near-field response from a point $\mathbf{p}$ to an array with element positions $\{\mathbf{u}_n\}$ can be written as

$$\mathbf{a}_{NF}(\mathbf{p}) = \frac{1}{\sqrt{N}} \left[e^{-j\frac{2\pi}{\lambda} d_1(\mathbf{p})}, \dots, e^{-j\frac{2\pi}{\lambda} d_N(\mathbf{p})} \right]^T, \quad (5)$$

where $d_n(\mathbf{p}) = \|\mathbf{p} - \mathbf{u}_n\|$ denotes the distance between point $\mathbf{p}$ and the n -th array element, λ is the carrier wavelength, and $\mathbf{a}_{NF} \in \mathbb{C}^{N \times 1}$, with $N = N_t$ for the BS array and $N = M$ for the STAR-RIS. Therefore, both $\mathbf{H}_{BS}$ and $\mathbf{h}_{SU,k}$ exhibit position-dependent phase variations and spatial focusing effects, which are key characteristics of near-field STAR-RIS systems. Accordingly, the direct and cascaded channels can be expressed using geometric near-field models with a limited number of dominant paths. Subsequently, the analog codebook, denoted by $\mathcal{F}$, can be constructed by collecting all the available array responses in (5).

The received signal at user k is expressed as

$$y_k = \tilde{\mathbf{h}}_k^H \mathbf{F}_{RF} \mathbf{F}_{BB} \mathbf{s} + n_k, \quad (6)$$

where $n_k \sim \mathcal{CN}(0, \sigma^2)$ is the additive white Gaussian noise. Let $\mathbf{f}_i$ denote the i -th effective precoding vector corresponding to the i -th scheduled stream. The received signal-to-interference-plus-noise ratio (SINR) at user k is expressed as

$$\gamma_k = \frac{|\tilde{\mathbf{h}}_k^H \mathbf{f}_k|^2}{\sum_{i \neq k} |\tilde{\mathbf{h}}_k^H \mathbf{f}_i|^2 + \sigma^2}, \quad (7)$$

and the achievable rate of user k is

$$R_k = \log_2(1 + \gamma_k). \quad (8)$$

In dense user scenarios, users with similar angular and spatial characteristics may be associated with highly correlated

analog beams, which leads to severe beam conflicts and stronger inter-user interference after hybrid precoding. As a result, conventional top-gain scheduling tends to maximize sum-rate at the cost of fairness, while purely orthogonality-based scheduling may lose useful beamforming gain.

Let $\mathcal{S} \subseteq \mathcal{K}_c$ denote the scheduled-user subset with $|\mathcal{S}| = K$. Our objective is to determine $\mathcal{S}$ together with the hybrid precoders in order to balance spectral efficiency and fairness. We consider the following utility-maximization problem as

$$\max_{\mathcal{S}, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB}}} U(\{R_k\}_{k \in \mathcal{S}}) \quad (9a)$$

$$\text{s.t. } |\mathcal{S}| = K, \quad (9b)$$

$$\|\mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB}}\|_F^2 \leq P_{\text{max}}, \quad (9c)$$

$$|[\mathbf{F}_{\text{RF}}]_{m,n}| = \frac{1}{\sqrt{N_t}}, \quad \forall m, n, \quad (9d)$$

where $U(\{R_k\}_{k \in \mathcal{S}})$ denotes the proportional-fairness utility function defined in (12). In this work, the STAR-RIS transmission and reflection coefficient matrices Φ_{T} and Φ_{R} are assumed to be preconfigured according to the energy-splitting protocol in (3). Therefore, they are embedded in the effective channel $\tilde{\mathbf{h}}_k$ and are not optimized in the proposed CA-SUS-HBF algorithm. The optimization in (9) is combinatorial and highly nonconvex due to the coupled user scheduling, analog beam selection, and digital precoding. Exhaustive search is computationally prohibitive when the candidate-user pool is large. This motivates the development of a low-complexity conflict-aware user scheduling and HBF method.

III. PROPOSED CA-SUS-HBF ALGORITHM

This section presents the proposed conflict-aware semi-orthogonal user scheduling and HBF (CA-SUS-HBF) scheme. The main idea is to select a user subset that not only has strong effective channel gains, but also exhibits low beam conflict and favorable rate balance after hybrid precoding.

For each candidate user $k \in \mathcal{K}_c$, let g_k denote its effective channel strength metric. To quantify the beam-conflict tendency, we define a conflict score

$$c_k = \sum_{j \in \mathcal{C}_k} \omega_{k,j}, \quad (10)$$

where $\mathcal{C}_k$ is the set of neighboring candidate users that share highly correlated beams or effective channels with user k , and $\omega_{k,j}$ is a nonnegative weight reflecting the severity of the conflict between users k and j .

Based on the gain and conflict characteristics, we define a priority metric for candidate screening as

$$\eta_k = \alpha_g g_k - \alpha_c c_k + \alpha_o \xi_k, \quad (11)$$

where ξ_k is an orthogonality-related term, while α_g , α_c , and α_o are nonnegative coefficients. A larger η_k indicates that user k is more favorable for scheduling.

After obtaining a tentative scheduled set and the corresponding hybrid beamformers, we evaluate the resulting rate vector $\mathbf{r} = [R_1, \dots, R_K]^T$. To balance throughput and fairness, we

adopt a utility function that combines the sum-rate and a proportional-fairness term such that

$$U(\mathbf{r}) = \alpha \sum_{k=1}^K R_k + (1 - \alpha) \sum_{k=1}^K \log(R_k + \epsilon) - \lambda_q \Psi(\mathbf{r}), \quad (12)$$

where $\alpha \in [0, 1]$ is the throughput-fairness tradeoff coefficient, $\epsilon > 0$ is a small constant, $\lambda_q \geq 0$ is a penalty weight controlling the influence of the quality-of-service (QoS) term, and $\Psi(\mathbf{r})$ is a QoS penalty term used to discourage overly small user rates. One simple choice is

$$\Psi(\mathbf{r}) = \sum_{k=1}^K \max(0, R_{\text{th}} - R_k), \quad (13)$$

where R_{th} is a target minimum rate.

The role of (12) is twofold. First, the sum-rate term preserves the throughput advantage of gain-based scheduling. Second, the logarithmic fairness term and QoS penalty reduce the possibility of starving weak users, which is essential in dense near-field STAR-RIS scenarios. Since the STAR-RIS coefficients are fixed during the considered scheduling interval, the proposed algorithm operates on the effective channels $\{\tilde{\mathbf{h}}_k\}$, which already include the direct and STAR-RIS-assisted propagation components.

The proposed CA-SUS-HBF algorithm consists of four stages. First, all candidate users are evaluated according to the priority metric in (11). A reduced candidate list is then formed by keeping the most promising users. Second, users are selected sequentially in a semi-orthogonal manner. At each iteration, the scheduler prefers a user with a large priority metric while avoiding severe beam conflicts with already selected users. This yields an initial scheduled set. Third, for the selected users, the analog beam indices are assigned from the predefined beam codebook, and the digital precoder is constructed accordingly. A local refinement step is further applied to improve the utility in (12) by testing limited user swaps and beam updates. Finally, a lightweight tail-repair mechanism is performed. If one scheduled user experiences a very small rate, the algorithm attempts to replace that user with another candidate that improves the utility without causing a large sum-rate loss. The proposed design avoids exhaustive search over all user subsets and beam combinations, while still capturing the most critical factors in dense user environments, namely beam conflict, channel orthogonality, and fairness-aware utility optimization.

The utility parameters in (12) are selected by an offline validation-based sweep over a predefined parameter grid. The final parameter set is chosen to balance average sum-rate, 5th-percentile rate, and Jain's fairness index, and is then fixed for all simulation results. The proposed CA-SUS-HBF avoids exhaustive search over all user subsets and beam assignments. Its main operations include pairwise conflict-aware metric evaluation, reduced-list sorting, sequential semi-orthogonal user selection, and a limited local refinement stage. As a result, the overall complexity grows polynomially with the candidate-user pool size, whereas exhaustive search scales combinatorially as $\binom{K_c}{K}$.

Algorithm 1: Proposed CA-SUS-HBF Algorithm

Input: $\mathcal{K}_c, \{\mathbf{h}_k(\Phi_{s_k})\}_{k \in \mathcal{K}_c}, \mathcal{F}, K$
Output: $\mathcal{S}, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB}}$

- 1 Compute $\{g_k, c_k, \eta_k\}_{k \in \mathcal{K}_c}$ using (10) and (11);
- 2 Construct reduced candidate list $\mathcal{L}$ by sorting η_k in descending order;
- 3 Initialize $\mathcal{S} \leftarrow \emptyset$;
- 4 **while** $|\mathcal{S}| < K$ **do**
- 5 Select $k^* = \arg \max_{k \in \mathcal{L} \setminus \mathcal{S}} \Gamma(k, \mathcal{S})$, where $\Gamma(\cdot)$ denotes the semi-orthogonal conflict-aware selection rule;
- 6 Update $\mathcal{S} \leftarrow \mathcal{S} \cup \{k^*\}$;
- 7 **end**
- 8 Construct $\mathbf{F}_{\text{RF}}$ from $\mathcal{F}$ for users in $\mathcal{S}$;
- 9 Obtain $\mathbf{F}_{\text{BB}}$ from the effective scheduled-user channel matrix;
- 10 Compute $\mathbf{r} = [R_1, \dots, R_K]^T$ using (7) and (8);
- 11 Evaluate $U(\mathbf{r})$ using (12);
- 12 **for** $i = 1, \dots, N_{\text{ref}}$ **do**
- 13 Generate a local update $(\mathcal{S}', \mathbf{F}'_{\text{RF}}, \mathbf{F}'_{\text{BB}})$;
- 14 Compute $\mathbf{r}'$ and $U(\mathbf{r}')$;
- 15 **if** $U(\mathbf{r}') > U(\mathbf{r})$ **then**
- 16 $\mathcal{S} \leftarrow \mathcal{S}', \mathbf{F}_{\text{RF}} \leftarrow \mathbf{F}'_{\text{RF}}, \mathbf{F}_{\text{BB}} \leftarrow \mathbf{F}'_{\text{BB}}, \mathbf{r} \leftarrow \mathbf{r}'$;
- 17 **end**
- 18 **end**
- 19 **if** $\min(\mathbf{r}) < R_{\text{th}}$ **then**
- 20 Perform tail repair by replacing the weakest user with a candidate maximizing (12);
- 21 **end**
- 22 **return** $\mathcal{S}, \mathbf{F}_{\text{RF}}, \mathbf{F}_{\text{BB}}$;

IV. SIMULATION RESULTS

This section performs system-level simulations to illustrate the performance of the proposed CA-SUS-HBF scheme in a near-field STAR-RIS-assisted multi-user mmWave downlink system. The main objective is to examine the throughput-fairness tradeoff achieved by the proposed method under different signal-to-noise ratio (SNR) levels, candidate-user pool sizes, and user-clustering conditions. In particular, the simulations focus on scenarios where beam conflicts become severe due to dense user distributions and limited RF resources.

We compare the proposed CA-SUS-HBF scheme with several representative benchmark methods. First, the FD-NF-RZF Upper Bound corresponds to a fully digital near-field transmission scheme with regularized zero-forcing (RZF) precoding, which serves as a useful performance upper bound for practical HBF designs since it is not restricted by analog beam selection [9], [11]. Second, the NF Top-Gain HBF benchmark schedules users mainly according to their effective channel gains and then performs near-field HBF. This scheme provides a throughput-oriented reference, but it does not explicitly account for beam conflicts or fairness [5], [12]. Third, the NF Semi-Orthogonal HBF benchmark combines semi-orthogonal user selection with near-field HBF, thereby reducing inter-user correlation, although it may lose beamforming gain in highly clustered scenarios [9], [13]. Fourth, the FF Approx. HBF benchmark applies a far-field approximation to the HBF design and is included to highlight the performance loss when near-field characteristics are neglected in STAR-

RIS-assisted systems [14], [15]. Finally, the Random HBF benchmark randomly selects users and beam assignments prior to digital precoding, providing a simple but useful baseline for illustrating the advantage of structured scheduling and conflict-aware design [5], [12].

Unless otherwise stated, the system parameters are set as follows. The carrier frequency is $f_c = 28$ GHz, corresponding to wavelength $\lambda = 3 \times 10^8 / f_c$. Half-wavelength spacing is adopted for both the BS antenna array and the STAR-RIS elements. The BS is equipped with $N_t = 16$ antennas and $N_{\text{RF}} = 4$ RF chains, while the STAR-RIS is modeled as an 8×8 planar surface with $M = 64$ passive elements. The BS and STAR-RIS centers are located at $(-12, 0, 8)$ m and $(0, 0, 6)$ m, respectively. The default candidate-user pool contains $K_c = 16$ users, with 8 users on the transmission side and 8 users on the reflection side. The maximum number of scheduled streams is $K = 4$. The transmission-side users are generated within $x \in [4, 14]$ m, while the reflection-side users are generated within $x \in [-14, -4]$ m. For both sides, the horizontal coordinate satisfies $y \in [-8, 8]$ m and the nominal user height is 1.5 m.

The total transmit power is adjusted according to the target SNR, i.e., $P_{\text{max}} = 10^{\text{SNR}/10} \sigma^2$, where the noise variance is normalized as $\sigma^2 = 1$. The default SNR is 10 dB, while the SNR sweep ranges from -5 dB to 20 dB with a step size of 5 dB. The STAR-RIS transmission and reflection power-splitting factors are initialized as $\beta_{m,R}^2 = \beta_{m,T}^2 = 0.5, \forall m, s$. For near-field beam search, the codebook $\mathcal{F}$ is constructed using 25 angular grid points and 9 range grid points, while the far-field benchmark uses 41 angular grid points. The candidate-user sweep considers $K_c \in \{8, 12, 16, 20, 24, 28\}$, and the angular cluster spread sweep uses $\{0.03, 0.05, 0.08, 0.12, 0.18, 0.25\}$ rad. The SNR sweep, candidate-user sweep, and conflict-sensitivity sweep are averaged over 80, 60, and 60 random drops, respectively. The utility parameters in (12) are fixed according to an offline validation-based tuning procedure that jointly considers average sum-rate, 5th-percentile scheduled-user rate, and Jain's fairness index.

Fig. 2 illustrates the average sum-rate performance of the considered schemes as the SNR increases from -5 dB to 20 dB. As expected, the fully digital FD-NF-RZF scheme provides the highest performance and serves as a useful upper bound. Among the practical schemes, the proposed CA-SUS-HBF achieves the highest sum-rate over the entire SNR range. In particular, its gain over NF Top-Gain HBF and NF Semi-Orthogonal HBF becomes more evident in the moderate-to-high SNR regime, where beam conflict and inter-user interference have a stronger impact on performance. The FF Approx. HBF benchmark suffers from the largest performance loss, which highlights the importance of accounting for near-field characteristics. Although the Random HBF benchmark attains relatively strong sum-rate in high SNR regimes, its performance remains less stable and does not benefit from structured conflict-aware scheduling, particularly in worst-off users, i.e., 5th-percentile performance shown in Fig. 3. Overall, Fig. 2 confirms that the proposed CA-SUS-HBF provides the

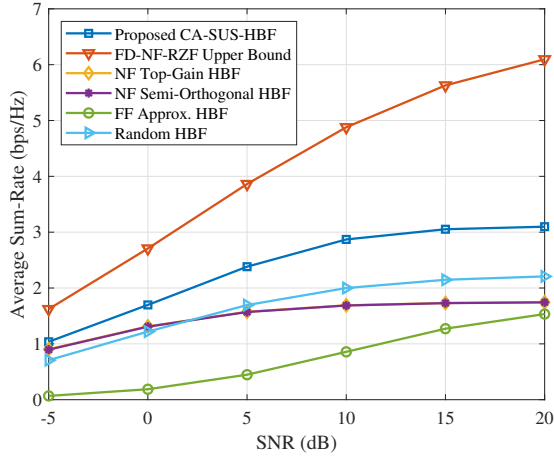


Figure 2. Average sum-rate versus SNR for different HBF schemes in the considered near-field STAR-RIS systems.

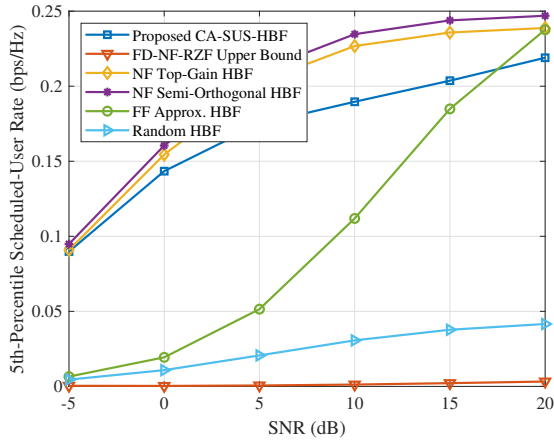


Figure 3. 5th-percentile scheduled-user rate versus SNR for different HBF schemes in the considered near-field STAR-RIS system.

most favorable sum-rate performance among the practical HBF schemes while maintaining a moderate gap to the fully digital upper bound.

Fig. 3 shows the 5th-percentile scheduled-user rate versus SNR, which reflects the tail performance among the scheduled users. The proposed CA-SUS-HBF consistently outperforms the Random HBF baseline and maintains a steadily increasing tail rate as SNR grows, indicating that the proposed utility design preserves a meaningful level of rate balance. At the same time, NF Top-Gain HBF and NF Semi-Orthogonal HBF achieve higher 5th-percentile rates in most operating points, since these schemes are more conservative in user selection and tend to produce more balanced scheduled-user sets. Therefore, Fig. 3 highlights the throughput–fairness tradeoff of the proposed design: CA-SUS-HBF does not maximize worst-user performance, but achieves a better overall balance between sum-rate and tail-user performance than purely random scheduling.

Fig. 4 shows the average sum-rate as the number of candid-

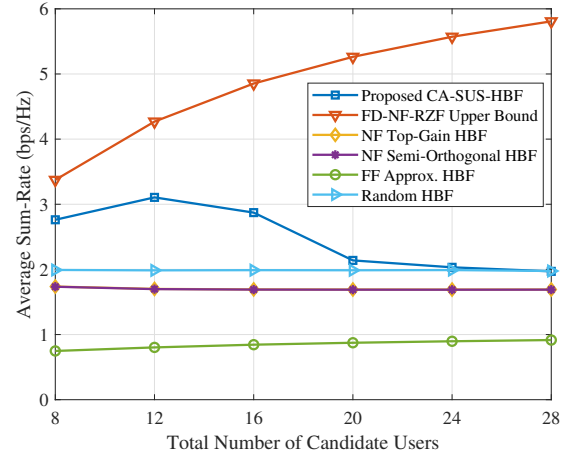


Figure 4. Average sum-rate versus the total number of candidate users.

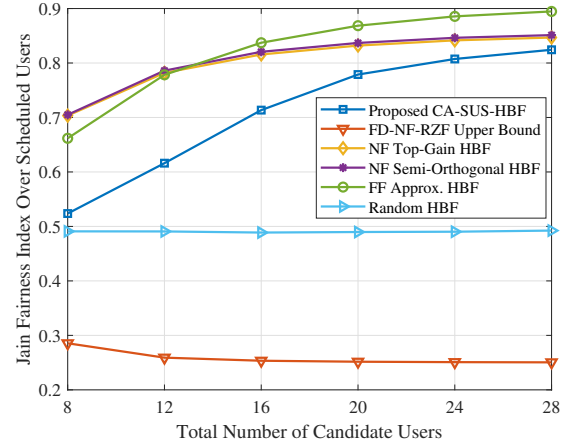


Figure 5. Jain's fairness index versus the total number of candidate users.

ate users increases. The FD-NF-RZF benchmark achieves the highest performance over the entire range, as expected, since it serves as a fully digital upper bound. Among the practical schemes, the proposed CA-SUS-HBF provides the highest sum-rate in the small-to-moderate user regime, indicating that the conflict-aware scheduling strategy can effectively exploit user diversity under limited RF chains. However, its sum-rate decreases when the candidate-user pool becomes large. This behavior is due to the fact that adding more users not only increases diversity, but also raises the probability of severe beam conflicts and channel overlap in the considered near-field STAR-RIS system. Under such conditions, the scheduler must increasingly trade off throughput against conflict mitigation and user-rate fairness, which leads to some sum-rate loss. Therefore, a larger candidate-user pool does not necessarily translate into higher sum-rate under practical HBF constraints.

Fig. 5 shows Jain's fairness index as the number of candidate users increases. It can be observed that the fairness of the proposed CA-SUS-HBF scheme improves steadily with the size of the candidate-user pool. This behavior indicates that a larger pool provides more flexibility for the scheduler to

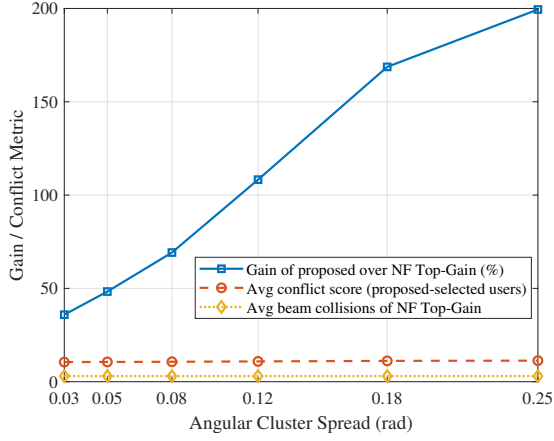


Figure 6. Performance gain and conflict-related metrics versus angular cluster spread.

select user subsets with better rate balance, thereby reducing extreme rate disparities among scheduled users. Nevertheless, the proposed scheme still achieves lower fairness compared with the NF Top-Gain and NF Semi-Orthogonal HBF benchmarks. This is mainly because CA-SUS-HBF explicitly incorporates conflict-aware prioritization and throughput-oriented selection, which may favor users with stronger effective channels or lower beam conflicts rather than strictly enforcing rate equalization. In contrast, the semi-orthogonal and gain-based schemes inherently promote more uniform user selection patterns. Overall, Fig. 5 highlights that the proposed method improves fairness as the system scales, while maintaining a deliberate tradeoff to preserve its sum-rate advantage under practical HBF constraints.

Fig. 6 illustrates the impact of angular cluster spread on the performance gain and conflict-related metrics. As the angular spread increases, the proposed CA-SUS-HBF scheme achieves a significantly higher sum-rate gain over the NF Top-Gain benchmark, demonstrating its effectiveness in handling spatially clustered user scenarios. This gain improvement indicates that the conflict-aware scheduling mechanism can better exploit spatial separability when user clusters become more distinguishable. At the same time, the average conflict score of the users selected by the proposed scheme remains relatively stable across different angular spreads, suggesting that the algorithm successfully controls beam conflicts even in challenging environments. In contrast, the average beam collision level of the NF Top-Gain scheme stays consistently low but does not translate into comparable throughput gains. Overall, Fig. 6 confirms that the proposed method effectively balances performance improvement and conflict mitigation, particularly in scenarios with pronounced angular clustering.

V. CONCLUSION

This paper proposed a conflict-aware semi-orthogonal user scheduling and HBF framework, termed CA-SUS-HBF, for near-field STAR-RIS-assisted multi-user systems. By jointly considering effective channel strength, beam conflict tendency,

and user-rate balance, the proposed method provides a practical low-complexity solution for user scheduling under HBF constraints. The numerical results showed that CA-SUS-HBF achieves a more favorable throughput–fairness tradeoff than conventional top-gain and semi-orthogonal baselines, while maintaining a reasonably small gap to the fully digital near-field upper bound. In particular, the proposed design was shown to be especially effective in dense user scenarios and under strong angular clustering, where beam conflicts become more severe. Future work will focus on further refining the utility design to improve the sum-rate, tail-user performance, and fairness tradeoff in dense-user regimes. In addition, the proposed framework can be extended to dynamic STAR-RIS scenarios with user mobility, channel aging, and time-varying configurations, where low-overhead learning-based scheduling may be useful for fast adaptation.

REFERENCES

- [1] R. Guirado, R. Verdecia-Peña, G. Perez-Palomino, E. Carrasco, and J. I. Alonso, "Prototype and performance assessment of mmWave active RIS-enhanced 6G wireless communications," *IEEE Trans. Green Commun. Netw.*, vol. 10, pp. 412–425, 2026.
- [2] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, 2019.
- [3] X. Mu, Y. Liu, L. Guo, J. Lin, and R. Schober, "Simultaneously transmitting and reflecting (STAR) RIS aided wireless communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 5, pp. 3083–3098, 2022.
- [4] W. Khalid, Z. Kaleem, R. Ullah, T. Van Chien, S. Noh, and H. Yu, "Simultaneous transmitting and reflecting-reconfigurable intelligent surface in 6G: Design guidelines and future perspectives," *IEEE Netw.*, vol. 37, no. 5, pp. 173–181, 2023.
- [5] O. E. Ayach, S. Rajagopal, S. Abu-Surra, Z. Pi, and R. W. Heath, "Spatially sparse precoding in millimeter wave MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 13, no. 3, pp. 1499–1513, Mar. 2014.
- [6] A. F. Molisch, V. V. Ratnam, S. Han, Z. Li, S. L. H. Nguyen, L. Li, and K. Haneda, "Hybrid beamforming for massive MIMO: A survey," *IEEE Commun. Mag.*, vol. 55, no. 9, pp. 134–141, 2017.
- [7] H. T. Dao and S. Kim, "Effective channel gain-based access point selection in cell-free massive MIMO systems," *IEEE Access*, vol. 8, pp. 108 127–108 132, 2020.
- [8] J. Mao, J. Gao, Y. Liu, and G. Xie, "Simplified semi-orthogonal user selection for MU-MIMO systems with ZFBF," *IEEE Wireless Commun. Lett.*, vol. 1, no. 1, pp. 42–45, 2012.
- [9] T. Yoo and A. Goldsmith, "On the optimality of multiantenna broadcast scheduling using zero-forcing beamforming," *IEEE J. Sel. Areas Commun.*, vol. 24, no. 3, pp. 528–541, 2006.
- [10] Q. Wu and R. Zhang, "Towards smart and reconfigurable environment: Intelligent reflecting surface aided wireless network," *IEEE Commun. Mag.*, vol. 58, no. 1, pp. 106–112, 2020.
- [11] C. Peel, B. Hochwald, and A. Swindlehurst, "A vector-perturbation technique for near-capacity multiantenna multiuser communication-part I: channel inversion and regularization," *IEEE Trans. Commun.*, vol. 53, no. 1, pp. 195–202, 2005.
- [12] A. Alkhateeb, O. El Ayach, G. Leus, and R. W. Heath, "Channel estimation and hybrid precoding for millimeter wave cellular systems," *IEEE J. Sel. Topics Signal Process.*, vol. 8, no. 5, pp. 831–846, 2014.
- [13] Z. Shen, R. Chen, J. Andrews, R. Heath, and B. Evans, "Low complexity user selection algorithms for multiuser MIMO systems with block diagonalization," *IEEE Trans. Signal Process.*, vol. 54, no. 9, pp. 3658–3663, 2006.
- [14] R. W. Heath, N. González-Prelcic, S. Rangan, W. Roh, and A. M. Sayeed, "An overview of signal processing techniques for millimeter wave MIMO systems," *IEEE J. Sel. Topics Signal Process.*, vol. 10, no. 3, pp. 436–453, 2016.
- [15] M. Cui and L. Dai, "Near-field wideband beamforming for extremely large antenna arrays," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 13 110–13 124, 2024.